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Data Sampling Strategies for Click Fraud Detection Using Imbalanced User Click Data of Online Advertising: An Empirical Review

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ABSTRACT
In the pay-per-click online advertisement model, fraudulent publishers' presence is rare than that of genuine publishers. This high-class imbalance between fraudulent and genuine publishers poses a challenge for the accurate classification of fraudsters due to the bias of automated learning models towards the outnumbered class. In this work, an empirical evaluation of significant popular data sampling methods is carried out using nine state-of-the-art learning models for classifying fraudsters in online advertisement. The main objective of this work is to understand the effect of oversampling, under-sampling, and hybrid sampling methods on the performance of various classifiers in click fraud detection. Extensive experiments are performed on the benchmark FDMA-2012 user-click dataset. The performance of each combination of data sampling method and classifiers is validated using average precision, recall, f1-score, and AUC. The results are also compared with the existing state-of-the-art models. The results suggest that adaptive synthetic sampling (ADASYN) oversampling with a gradient tree boosting (GTB) model performs best with an average precision score of 64.32%.

KEYWORDS
Class imbalance; Click fraud; Data sampling techniques; Ensemble learning; Online advertising; Pay per click

1. INTRODUCTION

Learning from the skewed data set has always been challenging in machine learning [1, 2]. The data sets of many real-world applications, including click fraud detection in online advertising [3–7], are highly skewed. Frequent browsing of websites by users tends to be an important revenue source for online advertising [8]. One such source of revenue for publishers is the pay-per-click (PPC) model of online advertising. But, the PPC model of online advertising is highly vulnerable to fraudulent activities due to enormous monetary involvement and may lead to fraudulent click activities at the publisher's end, which is known as click frauds. A challenging issue in detecting click fraud is the severe class imbalance ratio, genuine observations, and fraudulent observations. This extremely imbalanced distribution of class drives the designed model to be more biased towards the majority class. Dealing with skewed data involves techniques like algorithmic modification [9], cost-sensitive sampling [10, 11], ensemble learning methods [12–18], and data-level sampling methods [19–22]. Data sampling methods are highly versatile than the cost-sensitive and algorithmic sampling methods and can be used without a classifier. A higher depth of learning algorithms is required to modify the classification methods; the time needed to alter the learning algorithm is often higher than when using data-level methods for re-sampling the dataset.

This work aims to assess the effect of data-level sampling methods on classifiers' performance to achieve a reasonable balance of the number of observations in each class of severely imbalanced user-click dataset. The ratio of instances is re-sampled in the following ways to enhance classification: – (A) Oversampling the fraud (minority) instances using random oversampling (ROS), adaptive synthetic sampling (ADASYN), and support vector machine-based synthetic minority oversampling (SVM-SMOTE). (B) Under-sampling the non-fraud (majority) instances using random under-sampling (RUS), one-sided selection (OSS), and condensed nearest neighbor (CNN). (C) By oversampling and under-sampling the instances from fraud and non-fraud classes together with hybrid sampling methods, namely, synthetic minority oversampling with Tomek-link (SMOTETomek) and synthetic minority oversampling with the edited nearest-neighbor (SMOTEENN). The work focuses on skewness issues of a user-click multi-class dataset, rare in the literature of detecting click fraud for countering class imbalance. The following are the key contributions of the current research work:

- A comprehensive survey of studies conducted in the past 8 years to address the class imbalancing issues in FDMA 2012 user click dataset using data-level sampling strategies is performed.

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• The extensive empirical evaluation of popular eight data-level sampling methods with nine different learning models is performed for fraudulent publisher classification.

• The performance of models trained with datasets generated by different data balancing techniques is evaluated using average precision, sensitivity, F1-score and AUC metric.

• The results of earlier reported data sampling techniques and some new data sampling techniques are compiled for click fraud detection. Results are also compared with the other reported state-of-the-art work with different balancing techniques and learning models.

The rest of the paper is organized under the following sections. The related work is described in brief in Section 2. Section 3 discusses the materials and methods where the proposed framework with sampling techniques and evaluation measures are concerned. Experimental results are elaborately discussed in Section 4, while Section 5 concludes the work.

2. RELATED WORK

A fraud detection [23] is an area of data mining which comprises several methods to detect false activities in the Credit Card Fraud Detection [24, 25], Fraud Detection for E-commerce Transactions [26], Online auction fraud [27], and various other fields where illegitimate manipulations are possible. The current study focuses on evaluating the impact of data-level sampling strategies in the timely and accurate detection of fraudsters, we represent the work reported in the literature discussing the high-class imbalance issues encountered in the FDMA 2012 user click dataset towards click fraud detection in online advertising. One such case has been addressed by Perera *et al.* [6], where the presence of a few fraudulent publishers led to disproportionate instances in the classes, resulting in a skewed dataset. Such skewness in the dataset drives the predictive model biased to the over-represented class. Accordingly, distinct re-sampling methods, including up/downsampling and SMOTE, were used to address the class imbalance. They performed upsampling and downsampling by generating synthetic samples from the outnumbered class and removing samples randomly from the majority class until the dataset gets balanced. Similarly, Berrar [28] found data skewness a significant challenge to PPC internet advertising. The author examined severe class unbalances in the user-click dataset with the ratio of fraud, observation, and genuine instances as 72, 80, and 2929. The author combined the up/downsampling method with the random forest and grew each tree based on the re-balanced data and used the majority vote for prediction. Taneja *et al.* [29] found skewness in the dataset, having one class in majority while remaining in the minority. They addressed class imbalance utilizing SMOTE, which oversamples the minority class instances for balancing the dataset. They also used the Hellinger Distance Decision Tree (HDDT) classifier, which is insensitive to class imbalance. The classifier employs Hellinger distance as decision tree splitting criteria instead of using traditional splitting criteria, such as log-likelihood or entropy. They efficiently identified fraudsters utilizing SMOTE with the HDDT. Eventually, taking the advantage of benefits of GTB, Sisodia *et al.* [30] have addressed the severe issues confronted in the data analysis of online advertising, namely larger-sized data, complex structure, and skewness in class distribution. The author has assessed the GTB's effectiveness on FDMA, 2012 dataset towards identifying and classifying the fraudsters.

Researchers have shown that most of the re-sampling approaches for skewed classification are focused on oversampling the under-represented class utilizing SMOTE. Very few suites of methods are available for undersampling the outnumbered class. None of the studies evaluated the potential of hybrid sampling for detecting click fraud. The work evaluates data-level sampling methods' potential to address the disproportionate ratio of samples in the data, which hinders the classification methods' performance towards fraudsters classification.

3. MATERIALS AND METHODS

In this work, we carried out an experimental study towards highlighting the behavior of distinct data-level sampling strategies on a highly skewed raw user clicks dataset for re-sampling. The proposed framework is presented in Figure 1. We categorized this section into four parts. Section 3.1 elaborately discusses the characteristics of the dataset. The data-level sampling methods [31] used in the experiment are presented in Section 3.2. Finally, we discussed classification methods and evaluation measures in Sections 3.3 and 3.4, respectively.

3.1 Dataset Acquisition

Fraud Detection in Mobile Advertising (FDMA) 2012 competition data is used for experimentation in this study [32]. The dataset is available in Comma Separated Values (CSV) format.

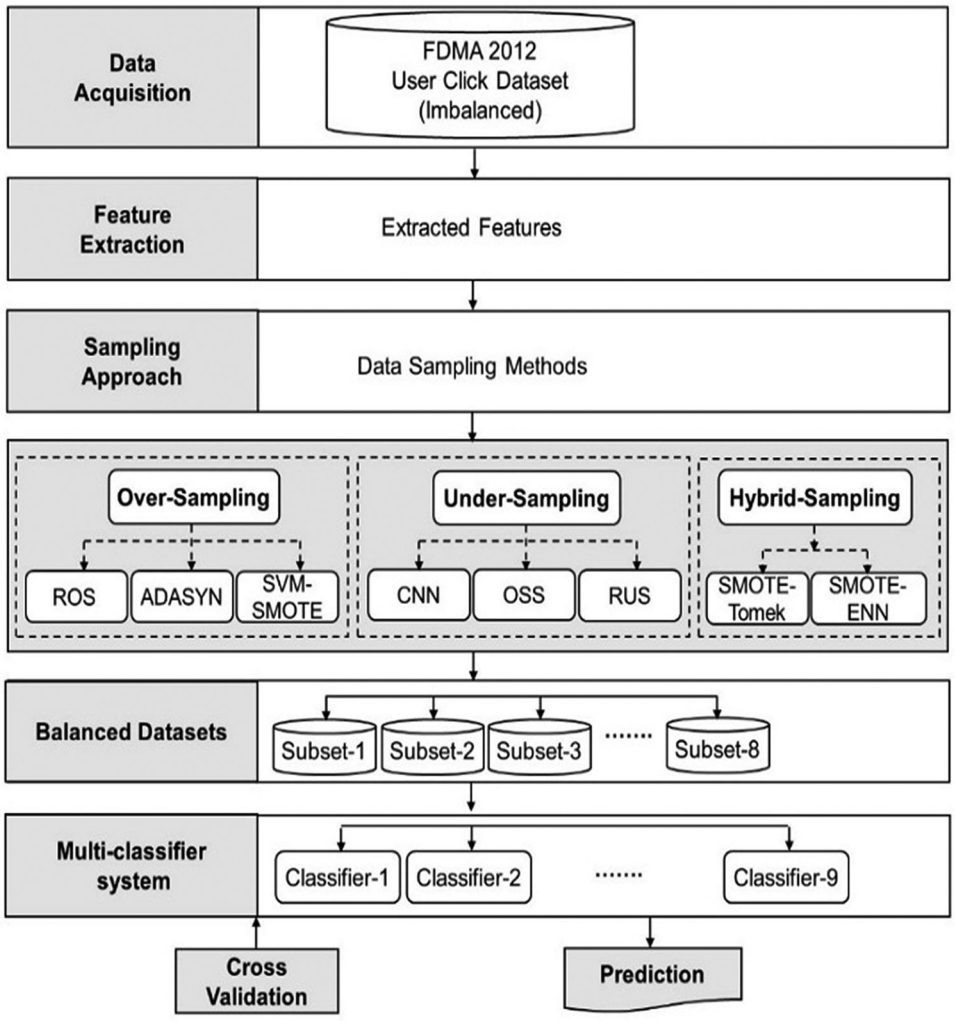

Figure 1: Visual illustration of the proposed framework

Table 1: Sample click statistics of the train set, validation set, and test set

Dataset	Id	Numericip	Deviceua	Pub id	Cam id	co	Timeat	Cat	Referer
Train clicks	13418094	3358683236	?	8ils9	8ggy8	ar	00:08.0	mc	?
	13418101	700994762	Nokia_N97	8i92t	8jqr2	dz	00:08.0	mc	0bc4mof2akrso8o0
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
Validation clicks	3038105	2088365076	?	8idhp	8kcce	th	01:24.0	mc	?
	3038107	2945070095	GT-S5360	8ixq4	8k2bq	my	01:24.0	ad	0odkolg7o60gss0g
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
Test clicks	9804629	3056950854	GT-S5360	dvry3	duhc8	th	07:37.0	mc	m47q9abkwwc87sz2
	9804631	3384765728	?	dv3va	dsfwc	cr	07:37.0	ad	?
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

The dataset comprises click dataset and publisher dataset, both of which fall into three different categories, collected at different time intervals. The entire statistics of the dataset are shown in Table 1. The click and publisher dataset samples are shown in Table 2 and Table 3, respectively. The click statistics shown in Tables 2 and 3 show that both datasets' sample includes missing values in some fields. The data presented in Table 2 shows that attributes "referer" and "deviceua" have missing values in the Train set, Validation set, and Test set, respectively.

Similarly, the attributes "bankaccount" and "address" comprise missing values in Table 2. Thus features need to be extracted to make meaningful representations from the raw data. For pre-processing, useful and relevant features are extracted from the FDMA 2012 user click dataset using several statistics, such as average and standard deviation. The raw user click dataset includes attributes in the alpha-numeric form that needs to be domain-specific for better data representation. A total of 103 features are extracted from

Table 2: Statistics of the dataset

			No. of publishers			
Dataset	Time period	No of clicks	Fraud	Observation	OK	Total
Train set	9–11 FEB 2012	1048575	72 (2.34%)	80 (2.60%)	2929 (95.07%)	3081
Validation set	23–25 FEB 2012	1048575	85 (2.77%)	84 (2.74%)	2895 (94.48%)	3064
Test set	8–10 MAR 2012	1048575	82 (2.73%)	71 (2.37%)	2847 (94.90%)	3000

Table 3: Sample click statistics of the publisher dataset

Dataset	Publisher	Bank account	Address	Status
Publisher 9–11 Feb12	8ikt9	0gftajxeyjfcc	4fj6vatj8cqoc	Fraud
	8ikpq	?	2z4toc2e5z0g	Observation
	⋮	⋮	⋮	⋮
Publisher 23–25 Feb12	8j0hb	25m0ecmqx00	35i52az4ufs4	OK
	8j0g9	?	4cy63ecu4ra	Fraud
	⋮	⋮	⋮	⋮
Publisher 8–10 Mar12	dv4bi	?	hxt8v20gxun	Observation
	dv48s	?	?	OK
	⋮	⋮	⋮	⋮

the original dataset to examine the behavior of the publisher.

3.2 Sampling Methods

A dataset is balanced when the class distribution is correct and identically presented in the same way [19]. Table 1 shows severe skewness in the dataset, where the number of genuine publishers exceeds fraudulent publishers and publishers' under-observation. This imbalanced nature of the dataset would make the predictive model inappropriate and biased, degrading machine learning algorithms' efficiency. The work in this experiment deals with skewness issues of a user click multi-class dataset by using the potential of the following data-level sampling methods:

3.2.1 Oversampling

The oversampling approaches operate over the instances of the minority class. For data balancing, these methods repeat instances of minority classes. In this paper, three techniques of oversampling, namely ROS [33], ADASYN [34], and SVM-SMOTE [35], are used to balance the dataset. ROS oversamples instances of the minority class by replicating the samples randomly with replacement. Instead of replicating instances, ADASYN sampling and SVM-SMOTE create new synthetic instances over the existing outnumbered examples.

3.2.2 Under-sampling

The under-sampling techniques work over the observations of the majority class. These methods alleviate or decrease instances of the majority class until the data get balanced. Three techniques, namely CNN [36], OSS [37, 38], and RUS [39], are employed in this experiment to under-sample the dataset. CNN [36] operates the 1-NN (One Nearest Neighbor) rule to under-sample the minority class instances, *i.e.* to minimize the duplicate samples. OSS alleviates noise by using the Tomeklinks. It also employs a 1-NN rule to determine which instances to be under-sampled. RUS balances the dataset by randomly reducing the cases of outnumbered class.

3.2.3 Hybrid Sampling

The hybrid sampling methods are a combination of both oversampling and under-sampling methods. Two-hybrid sampling techniques such as SMOTETomek [44] and SMOTEENN [45] are utilized to balance the user click dataset. In SMOTETomek, Tomeklinks acts as a data-cleaning tool and performs under-sampling of the majority instances when minority instances are oversampled using SMOTE. Likewise, in SMOTEENN, Edited Nearest Neighbor decreases those majority instances where the prediction made by nearest neighbor is found to be different from the outnumbered class.

3.3 Classification Methods

A total of nine state-of-the-art classification approaches employed in the experiment. K-Nearest Neighbor (KNN) [46, 47], Decision Tree (DT) [48, 49], Discriminant Analysis (DA) [50, 51], Logistic Regression (LR) [52–54], SVM [55–57], Gradient Tree Boosting (GTB) [30, 32, 58], AdaBoost (AdB) [59–61], Random Forest (RF) [32, 62], and Extra Trees (ET) [63, 64].

3.4 Assessment Methods for Multi-class Classification

While various approaches are available for assessing classification algorithms, accuracy and error rate cannot be considered standard evaluation metrics for handling class imbalance issues. This is because accuracy gets biased towards the majority class while ignoring the minority class. Since some of the measures are sensitive to the imbalanced data; thus, Table 4 demonstrates the following evaluation measures for multi-class classification problems that are appropriate for evaluating the imbalanced user click dataset. Other than the measures discussed in Table 4, we adopted AUC [40] metric for the valuable assessment of the performance in the context

Table 4: Classification measures for the imbalanced multi-class user-click data

Classification measures	Description	Equations	Eq. no.
Average precision (AP)	It is the proportion of correctly classified true examples/samples among the total actual predicted samples [40]. Due to its uniformity, it is deemed as a reliable measure for evaluating the imbalanced data [41]	$AP = \frac{\sum_{i=1}^{l} \frac{t_{p_i}}{t_{p_i}+f_{p_i}}}{l}$	(1)
Sensitivity (SE)	It is the measure of correctly classified true samples. It is suitable for evaluating imbalanced data as it depends on the true positive and false negative, which lies in the same column of the contingency table [42]	$TPR = \frac{t_{p_i}}{t_{p_i} + f_{n_i}}$	(2)
F-measure (F1)	It can be defined as the harmonic mean of precision and recall. Since this measure depends on sensitivity and specificity, it is well suitable with an imbalanced dataset [43]	$F1 = \frac{2t_{p_i}}{2t_{p_i} + f_{p_i} + f_{n_i}}$	(3)

Table 5: Tuned parameters of sampling methods

Methods	Name	Sampling strategy	Randomization	Size of the neighborhood
Oversampling	ROS	not majority	random state = 0	–
	ADASYN	not majority	random state = 0	n_neighbors = 4
	SVM-SMOTE	not majority	random state = 0	k_neighbors = 5
Under-sampling	CNN	not minority	random state = 0	n_neighbors = 1
	OSS	not minority	random state = 0	n_neighbors = 1
	RUS	not minority	random state = 0	–
Hybrid sampling	SMOTE-TOMEK	auto	random state = 0	–
	SMOTE-ENN	auto	random state = 0	–

Table 6: Number of instances before and after the sampling

Class		OK	Observation	Fraud	Total
Before sampling		2369 (93.90%)	76 (3.00%)	78 (3.10%)	2523
Oversampling	O1	2369 (33.30%)	2369 (33.30%)	2369 (33.30%)	7107
	O2	2369 (33.10%)	2392 (33.40%)	2391 (33.40%)	7152
	O3	2369 (36.80%)	2369 (36.80%)	1697 (26.40%)	6435
Under sampling	U1	366 (75.20%)	76 (15.60%)	45 (9.20%)	487
	U2	2267 (95%)	76 (3.20%)	44 (1.90%)	2387
	U3	76 (33.30%)	76 (33.30%)	76 (33.30%)	228
Hybrid sampling	H1	2357 (33.30%)	2363 (33.40%)	2363 (33.40%)	7083
	H2	1304 (22.00%)	2319 (39.10%)	2310 (38.90%)	5933

Note: O1: ROS, O2: ADASYN, O3: SVMSMOTE, U1: CNN, U2: OSS, U3: RUS, H1: SMOTE-TOMEK, H2: SMOTE-ENN.

of unbalanced datasets, which represents the capability of learning model towards distinguishing among classes.

4. EXPERIMENTAL RESULTS

The experiments utilized data-level sampling strategies implemented in Python version 3.7 using numpy [65], scipy [66], sklearn [67], imblearn [68], and pandas [69] libraries to assess the effectiveness of data-level sampling approaches for resolving class imbalance issues in user click dataset of online advertisement. This section shows the learning models' detailed results combined with various data sampling strategies to resolve the class imbalance issue. In this experimental work, the dataset is obtained from Fraud Detection in Mobile Advertising [32], a competition organized by the Buzz city Pvt. Ltd. Singapore. The user-click dataset is highly skewed as the percentage of samples of fraudulent publishers', publishers' under-observation (neither fraud nor genuine), and legitimate publishers are 2.77%, 2.74%, and 94.48%, respectively. The hitch with unbalanced distribution is that conventional machine learning approaches are biased towards the majority class, causing a higher misclassification rate. The details of the imbalanced user click multi-class dataset are summarized in Section 3.1.

The experiment is performed on the validation dataset. For training and testing we have adopted 10-fold cross-validation criterion. Three oversampling, three under-sampling, and two hybrid sampling methods are assessed to overcome the challenging issue of imbalance in the dataset. Table 5 illustrates the key parameters of sampling methods tuned for the better sampling of instances in this experiment. Table 6 presents the number of instances per class before and after the sampling.

Table 7 illustrates model parameters tuned in the experiment to achieve optimum classification efficiency. Nine individuals and ensemble classifiers are utilized in conjunction with data-level sampling methods to find a more robust classification solution. Our accessibility to fraud detection requires contemporary learners based on the features obtained from the validation dataset. Parameters of the classifiers are often tuned to ensure that their robust work is well generalized to the validation dataset.

Table 7: Tuned parameters of classifiers

S. No.	Classifiers	Tuned parameters
1.	LR	C = 1, random_state = 0
2.	SVM	C = 1, kernel = 'linear", degree = 3, gamma = 1, random_state = 0
3.	KNN	n_neighbors = 5, p = 2
4.	LDA	n_components = 2, solver = "svd"
5.	DT	criterion = "entropy", random_state = 0, max_depth = 3, min_samples_leaf = 5
6.	ET	n_estimators = 100, max_features = 50, criterion = "entropy", min_samples_split = 5, max_depth = 3, min_samples_leaf = 5, random_state = 0
7.	AB	n_estimators = 100, learning_rate = 1, random_state = 0
8.	RF	bootstrap = True, criterion = "entropy", max_depth = 3, n_estimators = 100, random_state = 0
9.	GTB	learning_rate = 0.1 (shrinkage), min_samples_split = 4, n_estimators = 100 (no. of trees) min_samples_leaf = 6, max_depth = 3, subsample = 0.3, random_state = 0

A comparative analysis of the performance of several data-level sampling methods on individual and ensemble methods is listed in Table 8. At first, to create a balanced dataset, oversampling methods, namely, ROS, ADASYN, and SVM-SMOTE, are utilized to oversample the minority instances. For evaluating the effectiveness of oversampling methods on the dataset, experiments were conducted using five individuals and four ensemble-based classifiers, where ADASYN indicates a good performance with an achieved precision score of 64.32%. Data from Table 8 clearly state the superior performance of ADASYN using the GTB among all the oversampling methods using – precision, recall, f1-score, and AUC. Second, three under-sampling methods, namely, OSS, CNN, and RUS, under-samples the majority of instances for re-balancing, where OSS (One-Sided Selection) outperforms the rest of the methods with an average precision score of 59.85%. Due to the loss of potentially useful information using under-sampling methods, GTB performs consistently poor. Its degraded performance is reported in Table 8. Two-hybrid sampling methods are then utilized to re-balance the class distribution of the user click dataset, namely, SMOTETomek, and SMOTEENN. Since hybrid sampling methods over-sample and under-sample, the instances simultaneously, they prevent the dataset from overfitting and losing useful information. SMOTEENN outperforms SMOTETomek with a precision score of 57.89%. Among all the classification methods, GTB yielded the best average precision score on the validation set, followed by the random forest in all the data-level sampling methods. Table 8 reports the

Table 8: Results of data-level sampling methods

Methods	Evaluation measures	Classification methods								
		LR	SVM	KNN	DT	LDA	AB	EXT	RF	GTB
ADASYN	AP	50.23	52.68	53.77	54.69	55.49	60.99	59.36	59.87	64.32
	SE	46.23	47.41	48.34	49.62	49.33	57.34	55.11	55.62	59.19
	F1	48.13	49.9	50.89	52.02	52.21	59.09	57.15	57.66	61.65
	AUC	58.35	59.28	59.93	62.47	63.22	67.61	65.70	64.86	72.44
SVM-SMOTE	AP	47.54	48.31	48.48	50.67	52.97	58.89	58.46	53.22	61.23
	SE	40.9	41.81	41.23	43.12	45.44	51.81	51.32	46.53	55.5
	F1	43.97	44.82	44.54	46.58	48.89	55.12	54.65	49.63	58.22
	AUC	54.36	56.54	56.92	57.61	59.48	65.75	65.13	61.54	68.47
ROS	AP	45.21	45.19	49.69	48.77	50.89	55.03	56.21	51.34	60.67
	SE	40.41	40.23	44.61	43.12	44.56	48.23	52.71	44.9	54.8
	F1	42.67	42.55	47.01	45.76	47.48	51.39	54.4	47.9	57.59
	AUC	53.94	54.14	58.54	57.23	59.68	62.85	64.31	60.24	67.52
OSS	AP	43.44	42.52	47.99	50.12	52.11	54.87	52.71	58.35	59.85
	SE	40.74	39.63	41.51	48.12	49.23	50.22	49.81	51.33	51.9
	F1	42.03	41.01	44.51	49.09	50.61	52.43	51.21	54.6	55.59
	AUC	52.89	52.42	54.64	58.43	60.12	61.45	60.78	65.38	66.67
CNN	AP	41.59	40.26	47.14	49.97	51.24	53.67	52.65	57.41	58.77
	SE	39.1	38.35	42.67	42.7	45.28	46.09	45.58	48.24	49.04
	F1	40.31	39.28	44.79	46.05	48.08	49.59	48.86	52.43	53.47
	AUC	50.26	48.56	53.47	57.65	58.81	62.51	60.24	66.12	67.54
RUS	AP	39.45	36.99	45.31	49.28	49.68	52.35	50.96	56.88	58.01
	SE	36.12	33.21	41.64	42.35	41.92	43.43	42.61	47.62	50.9
	F1	37.7	34.99	43.38	45.52	45.46	47.46	46.41	51.83	54.22
	AUC	46.21	45.36	53.45	56.18	57.41	59.37	59.68	64.17	66.17
SMOTEENN	AP	45.78	42.32	45.21	47.88	52.35	55.41	54.62	59.67	59.99
	SE	40.93	38.42	40.31	40.92	47.31	49.45	49.21	53.48	53.89
	F1	43.2	40.26	42.61	44.12	49.7	52.26	51.67	56.41	56.78
	AUC	52.06	51.56	52.14	54.67	60.39	62.97	59.42	67.57	66.54
SMOTETomek	AP	42.67	38.31	44.87	46.22	50.25	52.54	53.47	57.24	57.89
	SE	37.97	37.06	39.46	40.89	44.76	46.89	48.56	53.82	53.74
	F1	40.18	37.67	41.99	43.39	47.35	49.55	50.9	55.48	55.74
	AUC	52.14	45.33	52.95	53.21	56.46	58.34	61.85	64.27	65.13

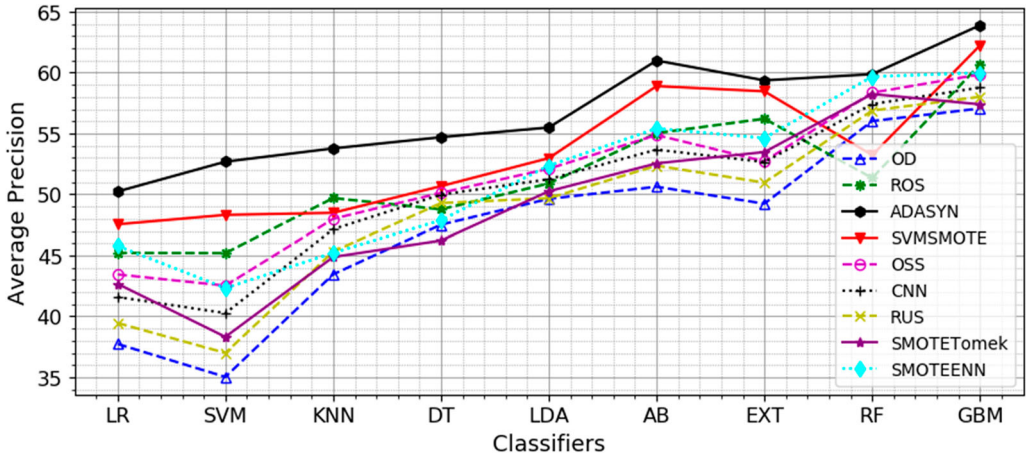

Figure 2: Evaluating the effectiveness of data-level sampling methods using the precision score

Table 9: Comparison of previous work with the current work utilizing data-level sampling methods on FDMA 2012 dataset

Author	Sampling method with classifier	Result
Taneja *et al.* [29]	SMOTE + HDDT	64.00% AP
Berrar [28]	Up/downsampling + RF	49.99% AP
Perera *et al.* [6]	SMOTE + RF	51.40% AP
Our Contribution	**ADASYN + GTB**	**64.32 % AP**

average precision score of 64.32%, 59.85%, and 59.99% by GTB using ADASYN, OSS, and SMOTEENN sampling methods. It has been observed from the results that ADASYN (oversampling approach) proves to be the right solution towards handling the imbalance issue of the dataset using the GTB classifier. Figure 2 illustrates the graphical presentation of data-level sampling methods' effectiveness. The x-axis represents classification methods, and the y-axis represents an average precision of classifiers obtained using the sampling methods. The graph demonstrates that the highest average precision score can be obtained using the ADASYN sampling method with the GTB classifier.

Table 9 represents a comparative study of the current work with previous literature to classify fraudulent publishers on FDMA 2012. Results demonstrate the promising performance of the ADASYN oversampling method using GTB with fine-tuned parameters that have outperformed the rest of the sampling methods with the highest average precision score towards fraudulent publisher's classification.

5. CONCLUSION

In this paper, a comprehensive empirical review of various popular data-level sampling strategies was performed on highly skewed user click data for click fraud detection. We reviewed the literature of the past 8 years on click fraud detection in online advertising and studied popular data sampling techniques and learning models. Data sampling techniques are applicable for fair balancing of the uneven class distribution of a dataset. Therefore, the work was focused on evaluating the impact of data-level sampling strategies, namely, oversampling (ROS, ADASYN, and SMOTE), under-sampling (RUS, OSS, and CNN), and hybrid sampling (SMOTETomek and SMOTEENN) techniques on the functioning of nine classifiers.

An extensive empirical review was performed to investigate the impact of the mentioned data sampling strategies on the classifiers' performance. Various parameters were tuned to compare the performance of the models. Precision, recall, f1-score, and AUC were used to calculate the classifiers' performance as the accuracy fails to measure the class imbalance problems. The empirical results suggested that the data oversampled using ADASYN with GTB has improved the classification result with an average precision score of 64.32%.

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